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3D curve weld seam path and posture planning based on line laser sensors

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ABSTRACT

The range of welding scenarios is expanding and becoming more complex in line with the development of manufacturing, which has led to increased demand for automatic and intelligent welding solutions. Line laser sensors are a crucial tool and technology in achieving intelligent welding. However, with the rise in the diversity of weld types and welding parts, the increasingly complex welded components pose challenges to point cloud construction. In contrast, the fixed posture of the welding torch in conventional digital welding makes it difficult to meet the welding needs of 3D curve weld seam. Therefore, this paper proposed a point cloud construction method based on robot pose. Firstly, the multi coordinate system transformation relationship was solved, and precise coordinate transformation between multi frame point clouds was achieved through coordinate transformation matrix, constructing a complex weld seam point cloud model. Furthermore, the point cloud was processed to extract weld seam information with the Ransac algorithm. Based on this, considering the characteristics of robot motion, the welding torch posture is divided into three components: deflection angle, elevation angle, and rotation angle, each of which is calculated separately to achieve welding posture planning. Experimental results have shown that the accuracy of the point cloud construction method proposed in this paper is better than 0.2mm, and the planning errors of the three posture angles are 0.75°, 1.2°, and 0.28°, which aligns well with the requirements of practical welding operations.

1. Introduction

In accordance with the manufacturing industry's continuous advancement, the applications and scenarios for welding have been on the rise. Welding has emerged as a significant approach to manufacturing in various fields, such as marine equipment [1,2], bridges [3–5], petrochemicals [6–8], transportation [9–12], etc. Traditional robot welding relies on the "teach-playback" mode, which is inefficient, cumbersome to operate, and difficult to adapt to the increasingly complex welding requirements in modern manufacturing [13]. As robot welding technology advances, the application of automation and intelligent welding is gradually increasing.

The use of visual sensors to construct models of weldments [14,15] and locate weld seams [16,17] is crucial in achieving automated welding. Scholars have conducted extensive research on this topic. Li et al. [18] proposed a surface reconstruction method for straight weld seams based on binocular structured light, achieving the construction of a single weld point cloud model. Chi et al. [19] used a depth camera to extract image features and point cloud recognition for straight weld seams, and proposed a method for robot pose estimation and 3D reconstruction of weldment. The accuracy of visual pose estimation was improved through 3D point cloud matching. Wang et al. [20] proposed a point cloud registration system for large planar components to achieve multi-view point cloud registration and proposed a multi-scale feature extraction network to reduce the impact of similar structures on registration. Wang et al. [21,22] proposed a method for constructing a multi-seam point cloud model based on photography with monocular cameras and laser rangefinders for circular weld seam of the tubesheet, and further proposed the voxel density method to process the point cloud, achieving multi-seam recognition. The above research mainly focuses on the construction of point cloud models for various types of welds, but intelligent welding requires recognition and positioning of weld seams. Xiao et al. [23] proposed a machine learning method based on RCNN to recognize weld types for six typical linear weld seams, and achieved initial point detection and localization of weld seams. Wang et al. [24] proposed a voxel based deep learning network for identifying

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Fig. 1. The experimental system of this paper (a) The overall structure of the system (b) Sensor installation position (c) Welding machine.

circular welds in point cloud models. Fang et al. [25] proposed a point cloud model construction method based on binocular vision for S-shaped and saddle shaped weld seams, and further carried out curve fitting to complete weld seam positioning.

The aforementioned research mainly concentrates on the construction of point cloud models, the recognition and positioning for planar weld seams. However, the welding scene also includes 3D weld seams. To ensure welding quality, the welding torch must maintain a constant angle with the weld seam during the welding process, resulting in differences in the posture of the welding torch at different positions of the 3D weld seam. Consequently, in addition to identifying and positioning the torch, it is necessary to calculate and plan the optimal welding posture at each position for 3D weld seams. Scholars have conducted relevant research on welding posture calculation. Yang et al. [26] designed a binocular structured light system for weld seams such as straight lines, Z-lines, and curves, obtained a point cloud model of the workpiece, and calculated the position and posture of the weld seam by extracting information from the weld groove. Wu et al. [27] planned the scanning path for corner joint curve weld seams based on the 3D CAD model, further collected the point cloud model of the weldment, and finally extracted the weld seam and implemented feature point pose calculation. Yang et al. [28] developed a binocular structured light system for corner, lap, and butt straight weld seams, and constructed a point cloud model for workpiece, and then performed point cloud segmentation to achieve path planning and pose estimation. However, the above research mainly focuses on planar weld seams, with minimal attention devoted to 3D curved weld seams, which means that pose calculation is relatively basic. Furthermore, Geng et al. [29] focused on spatial structural components, combined point cloud models with theoretical 3D models to extract weld seams, and completed welding pose calculation and path planning based on spatial structures. This method involves 3D weld path planning; however, the construction process of the point cloud model does not incorporate the robot pose, which consequently affects the accuracy of the point cloud model. Zou et al. [30] developed a laser sensor for 3D curved welds and extracted the weld trajectory frame by frame, further achieving the calculation of weld trajectory position and attitude angle. However, this method has only been tested for one type of weld seam and the positioning accuracy needs to be improved. In summary, the following issues mainly exist in the current weld path planning: firstly, the accuracy of complex weld point cloud models cannot be assured. Complex welding components require multiple point cloud splicing, and point cloud models cannot guarantee this. Secondly, the accuracy of 3D curve weld path planning must be enhanced.

In order to address the above issues, a 3D curve weld path and posture planning method based on line laser was proposed in this paper. The main contributions of this paper are as follows:

(1) A point cloud model construction method combining robot pose and line laser sensor was proposed. By using robot pose as the

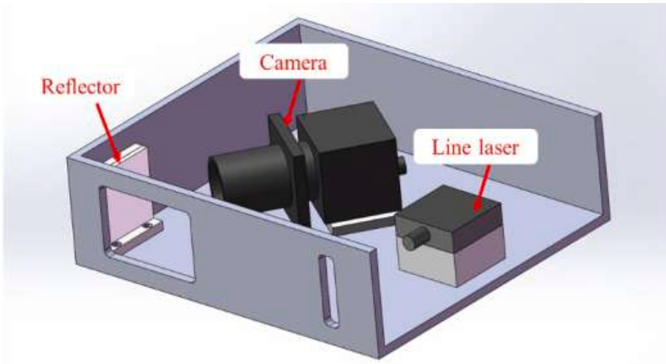

Fig. 2. The structure of laser vision sensor in this paper.

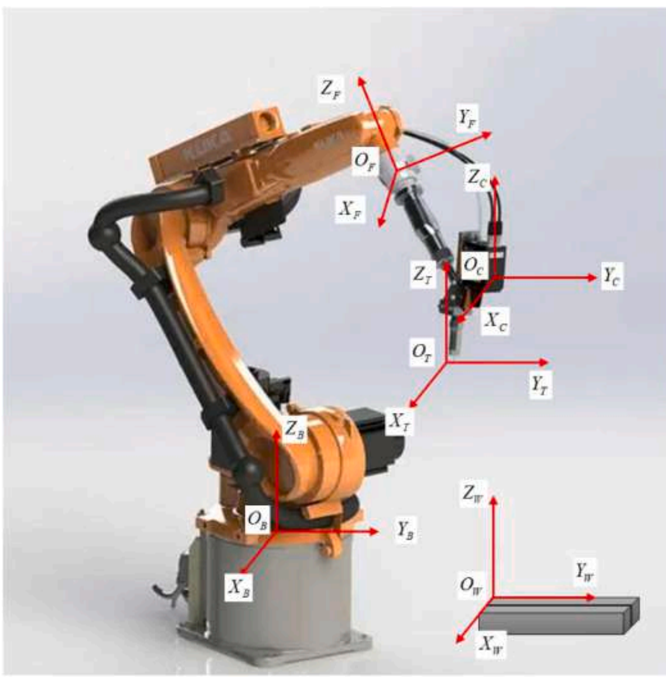

Fig. 3. The structure of laser vision sensor in this paper.

benchmark for point cloud stitching, the accuracy of point cloud model construction has been enhanced.

(2) A high-precision 3D curve welding seam pose calculation method has been proposed. By calculating the welding gun pose of the point cloud model frame by frame, high-precision pose planning is generated.

The following section of this paper is arranged as follows: The experimental platform and sensors are introduced in Section 2. The point cloud model construction method is introduced in Section 3. The weld seam path and posture are calculated in Section 4. The corresponding experimental verification and analysis are conducted in Section 5. The conclusions are given in Section 6.

2. Experimental system

Based on the characteristics of 3D curved weld seam welding, the robot experimental system has been built in this paper, as shown in Fig. 1. The experimental system mainly includes welding robot, laser vision sensor, welding machine, computer, etc. The laser vision sensor is installed in front of the welding torch, as shown in Fig 1 (b). The robot in this paper is Kuka KR 50 R2100, and the laser vision sensor is a self-developed sensor, as shown in Fig. 2. The data transmission between Kuka robots and computers is achieved through the TCP protocol, while the data transmission between laser vision sensors and computers is achieved through the USB 3.0 interface.

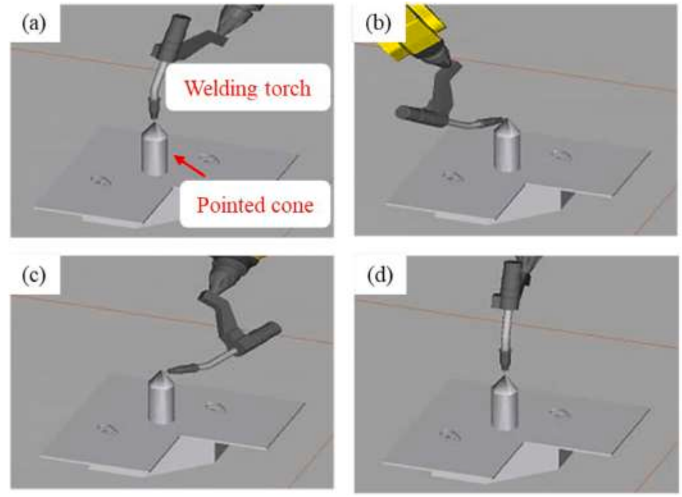

Fig. 4. Calibration between BCS and TCS.

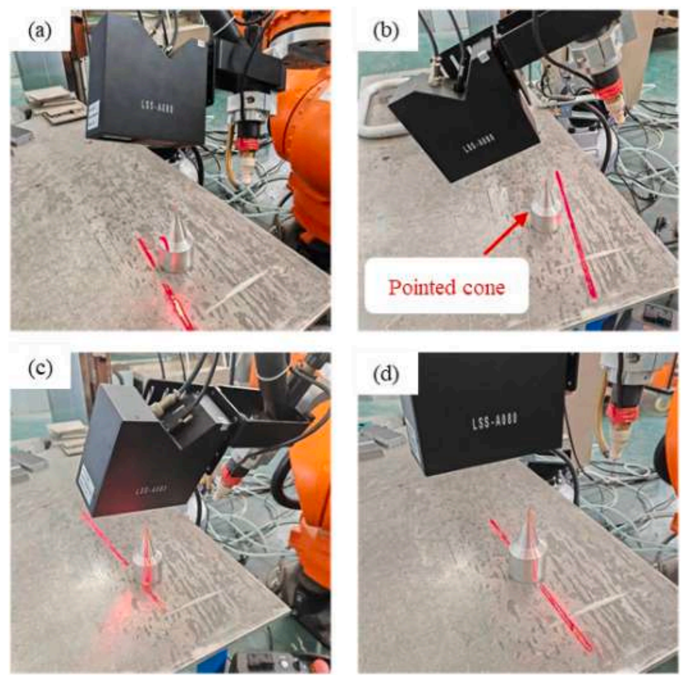

Fig. 5. Calibration between BCS and CCS.

3. Construction of 3D curve weld seam point cloud model

3.1. Robot multi-coordinate system conversion

The robot welding system designed in this paper employs a multitude of coordinate systems, including the robot base coordinate system (BCS) $O_B - X_B Y_B Z_B$, flange coordinate system (FCS) $O_F - X_F Y_F Z_F$, camera coordinate system (CCS) $O_C - X_C Y_C Z_C$, workpiece coordinate system (WCS) $O_W - X_W Y_W Z_W$ and welding torch coordinate system (TCS) $O_T - X_T Y_T Z_T$, as shown in Fig. 3. The robot has completed zero position correction, and the FCS position is the robot's coordinate in BCS. In order to accurately obtain the position of the workpiece and perform

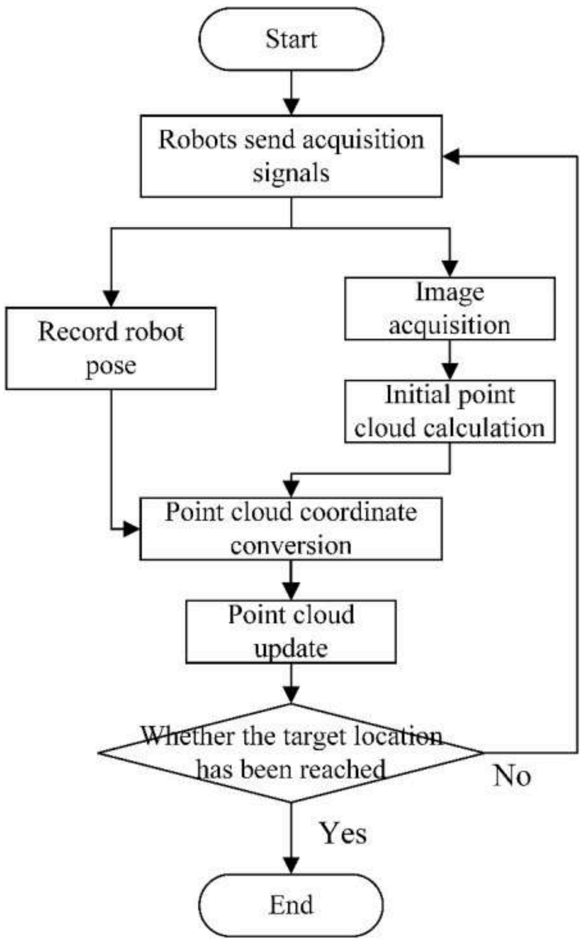

Fig. 6. Point cloud model construction process.

welding, it is necessary to obtain the accurate coordinate transformation relationship.

In this paper, the internal parameters of the camera and the external parameters between the camera and the line laser are determined values. As a result, the coordinate transformation matrix R_{wc} and translation vector t_{wc} of the sensor can be calculated. The conversion relationship between BCS and TCS can be obtained through the robot tool coordinate system calibration method, as shown in Fig. 4. A pointed cone was used as the calibration object, and the end of the welding torch was moved to reach the pointed cone in four different postures. The coordinate transformation matrix R_{tb} and translation vector t_{tb} between BCS and TCS can be obtained through these four postures. Through this method, TCS calibration can be completed, and TCS coordinates can be obtained in the robot.

Similarly, the coordinate transformation relationship between CCS and TCS can be obtained through robot hand eye calibration, as shown in Fig. 5. The pointed cone is selected as the calibration object, and the line laser sensor is controlled to move and capture the apex of the pointed cone. Repeating the above operation at four different positions, and defining the distance between the points mapped from the transformation matrix to the TCS and the real points as residuals, the objective function by minimizing the residuals is established:

$$\begin{bmatrix} x_t \\ y_t \\ z_t \\ 1 \end{bmatrix} = \begin{bmatrix} \mathrm{R}_{ct} & t_{\mathrm{ct}} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix} \tag{1}$$

$$^{T}_{C}\widehat{H} = \begin{bmatrix} \mathrm{R}_{ct} & t_{\mathrm{ct}} \\ 0 & 1 \end{bmatrix} \tag{2}$$

$$\min_{^{T}_{C}\widehat{H}} r\left(^{T}_{C}\widehat{H}\right)^2 = \sum_{i=1}^{4} \| \, ^{T}_{C}\widehat{H} P_i^C - P_i^T \, \|^2 \tag{3}$$

where P_i^C is the coordinate of a point in CCS, and P_i^T is the corresponding coordinate of the point in TCS. The precise solution of coordinate transformation matrix R_{ct} and translation vector t_{ct} between CCS and TCS can be iteratively achieved by using nonlinear least squares method [31]:

$$\begin{bmatrix} \mathrm{R}_{ct} & t_{\mathrm{ct}} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.018 & -0.999 & -0.001 & -64.57 \\ -1.033 & -0.016 & -0.022 & -1.020 \\ 0.031 & 0.001 & -1.034 & -0.637 \\ 0 & 0 & 0 & 1.000 \end{bmatrix} \tag{4}$$

Further verification shows that the calibration error is 0.035mm, which fully meets the practical welding requirements.

3.2. Point cloud model construction

By collecting data through laser vision sensors and performing image processing, local point clouds can be obtained. However, the determination of the coordinate relationship between multiple frame point clouds remains a challenge, thereby resulting in the inability to accurately acquire the complete point cloud model of the workpiece. To address this limitation, a point cloud model construction method based on robot pose was proposed in this paper, as shown in Fig. 6.

Firstly, after the robot moves to the target position, a collection signal is sent, and the robot pose is recorded while the laser vision sensor captures the image. During the collection process, the robot and sensor data will be recorded with timestamps to ensure data alignment. The data collection and robot pose recording are conducted at the same frequency. The laser vision sensor image can be processed to calculate the initial point cloud through data processing, which is in the CCS. The robot pose is a six dimensional vector:

$$[\mathrm{X} \quad Y \quad Z \quad A \quad B \quad C]^T \tag{5}$$

where $(X \quad \mathrm{Y} \quad Z)$ is the coordinate of the laser vision sensor under TCS, $(A \quad B \quad \mathrm{C})$ is the angle between TCS and BCS, A is the angle of rotation around Z_{T}, B is the angle of rotation around Y_T, and C is the angle of rotation around X_{T}. Due to the dynamic variation of rotation angle, the coordinate conversion relationship between TCS and BCS is not fixed and can be expressed as:

$$\mathrm{R}_{tb} = R(Z_T, A) R(Y_T, B) R(X_{\mathrm{T}}, C) \tag{6}$$

where $R(Z_T, A)$ is the rotation around Z_{T}, $R(Y_T, B)$ is the rotation around Y_{T}, and $R(X_T, C)$ is the rotation around X_{T}. Furthermore, the above rotations can be solved by the following formula:

$$R(Z_T, A) = \begin{bmatrix} \cos A & -\sin A & 0 \\ \sin A & \cos A & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{7}$$

$$R(Y_{\mathrm{T}}, B) = \begin{bmatrix} \cos B & 0 & \sin B \\ 0 & 1 & 0 \\ -\sin B & 0 & \cos B \end{bmatrix} \tag{8}$$

$$R(X_T, C) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos C & -\sin C \\ 0 & \sin C & \cos C \end{bmatrix} \tag{9}$$

Meanwhile, the translation vector is as follows:

$$t_{tb} = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \tag{10}$$

By combining formulas (6)–(10), TCS can be converted to BCS to achieve point cloud coordinate conversion;

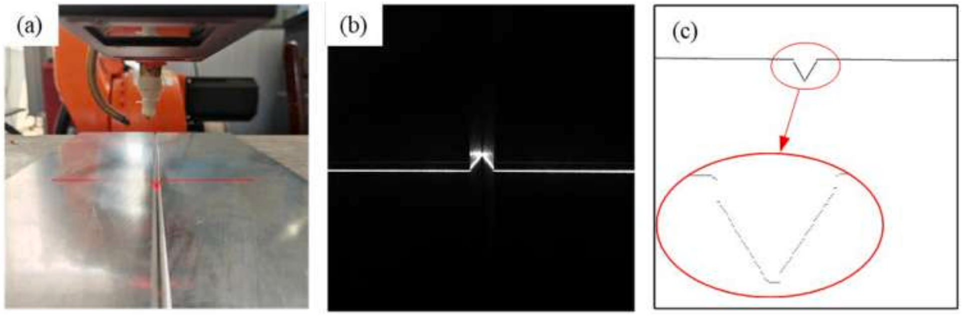

Fig. 7. Single frame point cloud model obtained by line laser sensor (a) The construction process of point cloud models (b) Images captured by line laser sensors (c) Single frame point cloud.

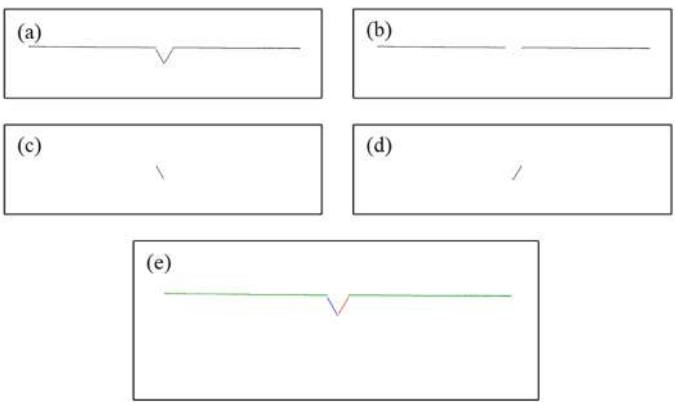

Fig. 8. RANSAC point cloud line extraction (a) Raw point cloud data (b) The first line in point cloud (c) The second line in point cloud (d) The third line in point cloud (e) Multi line extraction from point clouds.

$$\begin{bmatrix} x_b \\ y_b \\ z_b \\ 1 \end{bmatrix} = \begin{bmatrix} R_{tb} & t_{tb} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_t \\ y_t \\ z_t \\ 1 \end{bmatrix} \tag{11}$$

Furthermore, point cloud data from CCS can be converted into BCS:

$$\begin{bmatrix} x_b \\ y_b \\ z_b \\ 1 \end{bmatrix} = \begin{bmatrix} R_{tb} & t_{tb} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R_{ct} & t_{ct} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix} \tag{12}$$

Each frame of point cloud corresponds to a robot pose through the timestamp, and the point clouds between different frames are converted into BCS to achieve point cloud stitching by combining formulas (4) and (12). By repeating the above process until data collection is completed, the construction of the workpiece point cloud model can be achieved. The coordinate system of the constructed point cloud model is BCS.

4. 3D curve weld seam path and posture planning

4.1. Weld seam point cloud path planning

In Section 3, the construction of the point cloud model for the workpiece was completed. To move forward, it is necessary to process the point cloud to obtain welding seam information. The point cloud model obtained by the line laser sensor consists of multiple frames of data and the single frame point cloud data as shown in Fig. 7. It can be seen that the single frame point cloud data is relatively regular, presenting obvious corners at the weld seam groove.

The position of the weld seam is at the intersection of the groove. Considering the characteristics of single frame point cloud data, a data processing method based on RANSAC is proposed in this paper. The point cloud can be divided into inner and outer points by the RANSAC algorithm, and the first line can be obtained by fitting the inner points. Furthermore, by using the RANSAC again to process the outer points and re-dividing them into inner and outer points, the second line can be obtained. Repeating the above process, the extraction of multiple lines from the point cloud is completed, as shown in Fig. 8. It can be seen that the RANSAC algorithm accurately extracted multiple lines, effectively avoiding the interference of noise in Fig. 7 (c). Each line can be expressed as follows:

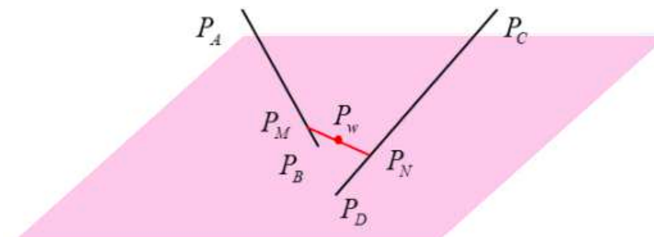

Fig. 9. The spatial position relationship of the fitted line.

$$\begin{cases} x = az + b \\ y = cz + d \end{cases} \tag{13}$$

Through the least squares algorithm, it can be calculated that:

$$a = \frac{n\sum_{i=1}^{s} x_i z_i - \sum_{i=1}^{s} x_i \sum_{i=1}^{s} z_i}{n\sum_{i=1}^{s} z_i^2 - \sum_{i=1}^{s} z_i \sum_{i=1}^{s} z_i} \tag{14}$$

$$b = \frac{\sum_{i=1}^{s} x_i - a\sum_{i=1}^{s} z_i}{n} \tag{15}$$

$$c = \frac{n\sum_{i=1}^{s} y_i z_i - \sum_{i=1}^{s} y_i \sum_{i=1}^{s} z_i}{n\sum_{i=1}^{s} z_i^2 - \sum_{i=1}^{s} z_i \sum_{i=1}^{s} z_i} \tag{16}$$

$$d = \frac{\sum_{i=1}^{s} y_i - c\sum_{i=1}^{s} z_i}{n} \tag{17}$$

where $(x_i \quad y_i \quad z_i)$ is the coordinate of point in each segment of the point cloud.

The equations for each segment of the line can be obtained through formulas (13)–(17). Due to fitting errors, the equations of each segment cannot be assured to be coplanar, as shown in Fig. 9, and further calculation of the weld position is needed. Assuming two lines are $P_A P_B$ and $P_C P_D$, each point can be represented as:

$$\begin{cases} P_A(x_a, y_a, z_a) \\ P_B(x_b, y_b, z_b) \end{cases} \tag{18}$$

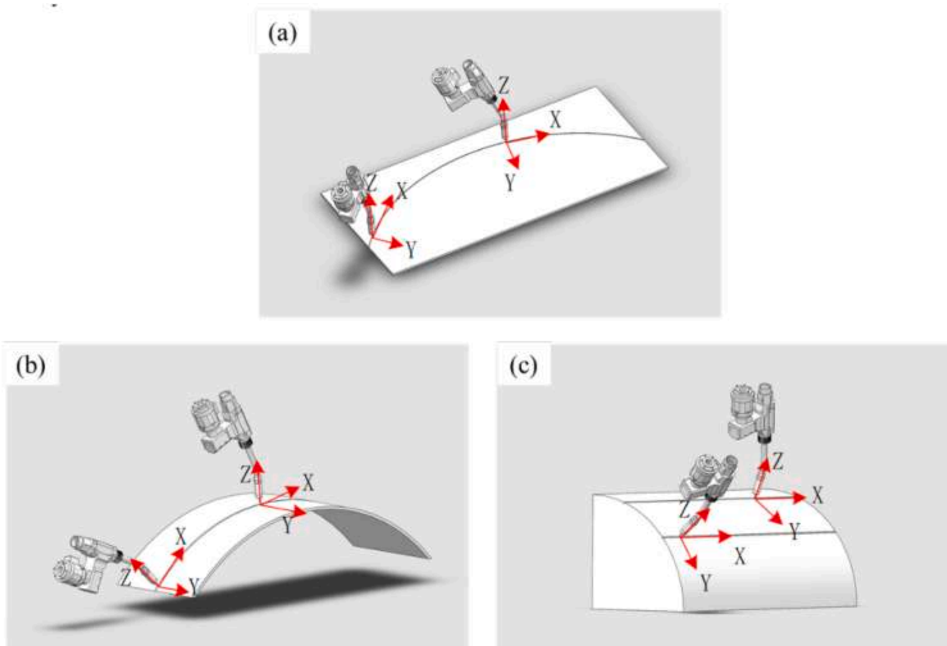

Fig. 10. Classification of welding posture (a) Deflection angle adjustment (b) Elevation angle adjustment (c) Rotation angle adjustment.

$$\begin{cases} P_C(x_c, y_c, z_c) \\ P_D(x_d, y_d, z_d) \end{cases} \tag{19}$$

Assuming that $P_M P_N$ is the common perpendicular of $P_A P_B$ and $P_C P_D$, then:

$$\begin{cases} P_M(x_m, y_m, z_m) \\ P_N(x_n, y_n, z_n) \end{cases} \tag{20}$$

$$P_A P_M = t_1 P_A P_B \tag{21}$$

$$P_C P_N = t_2 P_C P_D \tag{22}$$

where t_1 and t_2 are constants. According to formulas (18)–(22), the coordinates of P_M and P_N are:

$$\begin{cases} x_m = t_1(x_b - x_a) + x_a \\ y_m = t_1(y_b - y_a) + y_a \\ z_m = t_1(z_b - z_a) + z_a \end{cases} \tag{23}$$

$$\begin{cases} x_n = t_2(x_d - x_c) + x_c \\ y_n = t_2(y_d - y_c) + y_c \\ z_n = t_2(z_d - z_c) + z_c \end{cases} \tag{24}$$

Since $P_M P_N$ is perpendicular to $P_A P_B$ and $P_C P_D$, then:

$$F_1(a,b) = [(x_b - x_a)(x_b - x_a) + (y_b - y_a)(y_b - y_a) + (z_b - z_a)(z_b - z_a)] \tag{25}$$

$$F_1(c,d) = [(x_d - x_c)(x_d - x_c) + (y_d - y_c)(y_d - y_c) + (z_d - z_c)(z_d - z_c)] \tag{26}$$

$$F_2(a,b) = [(x_b - x_a)(x_d - x_c) + (y_b - y_a)(y_d - y_c) + (z_b - z_a)(z_d - z_c)] \tag{27}$$

$$F_3(a,b) = [(x_b - x_a)(x_c - x_a) + (y_b - y_a)(y_c - y_a) + (z_b - z_a)(z_c - z_a)] \tag{28}$$

$$F_3(c,d) = [(x_d - x_c)(x_c - x_a) + (y_d - y_c)(y_c - y_a) + (z_d - z_c)(z_c - z_a)] \tag{29}$$

$$t_2 F_2(a,b) - t_1 F_1(a,b) + F_3(a,b) = 0 \tag{30}$$

$$t_2 F_1(c,d) - t_1 F_2(a,b) + F_3(c,d)(c,d) = 0 \tag{31}$$

Assuming $P_w(x_w, y_w, z_w)$ is the weld point, then:

$$\begin{cases} x_w = \dfrac{x_m + x_n}{2} \\ y_w = \dfrac{y_m + y_n}{2} \\ z_w = \dfrac{z_m + z_n}{2} \end{cases} \tag{32}$$

By combining formulas (23)–(32), the position of the weld seam can be solved. By repeating the above process, the welding points corresponding to the complete point cloud can be obtained. Assuming the obtained point set is P:

$$P = \{P_1, P_2, ..., P_n\} \tag{33}$$

The points in P may have fluctuations, which may lead to unstable robot motion. Therefore, NURBS is selected to smooth the points [32]. The smoothed curve is:

$$C(u) = \frac{\sum_{i=0}^{n} \omega_i P_i N_{i,k}(u)}{\sum_{i=0}^{n} \omega_i N_{i,k}(u)} \tag{34}$$

where ω_i is the weight, $N_{i,k}(u)$ is the B-Spline basis function, and $C(u)$ is the point on the curve. Thus, the robot path planning is completed.

4.2. Curve weld seam posture planning

The position of the weld seam is obtained in Section 4.1. However, planning the welding posture for 3D curved weld seams is still required. Considering the motion mode of the robot, when adjusting the welding posture, the posture can be decomposed into α, β and γ adjustment:

$$R_{xyz}(\alpha, \beta, \gamma) = R(Z, \alpha) R(Y, \beta) R(X, \gamma) \tag{35}$$

where $R(Z, \alpha)$ is the deflection angle, $R(Y, \beta)$ is the elevation angle, and $R(X, \gamma)$ is the rotation angle, as shown in Fig. 10. It can be seen that the deflection angle α is the angle of rotation around the Z-axis, the elevation angle β is the angle of rotation around the Y-axis, and the rotation

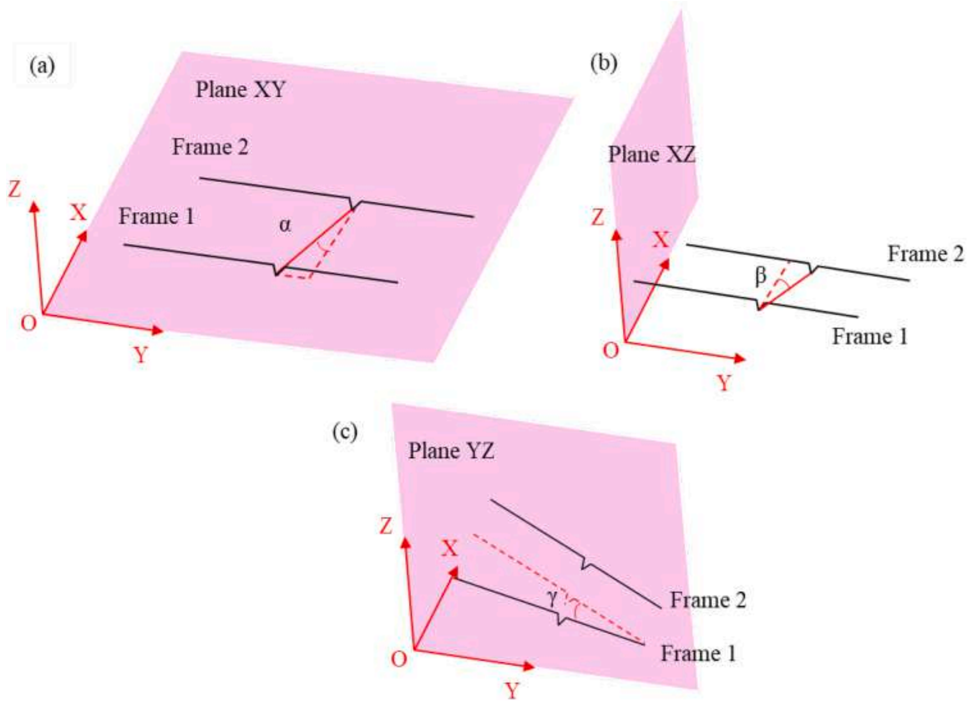

Fig. 11. Welding posture calculation (a) Deflection angle calculation (b) Elevation angle calculation (c) Rotation angle calculation.

angle γ is the angle of rotation around the X-axis. For industrial robots, all motion postures can be represented by the three angles mentioned above, including lines, curves, spatial curves, and other motions [33]. In the scene of complex movements such as spatial curves, three angles need to be adjusted simultaneously to achieve flexible and accurate robot movements.

Based on the position relationship of the weld seam between multiple point clouds, the corresponding posture of the welding torch can be calculated. The calculation of deflection angle is shown in Fig. 11 (a). Two adjacent frames of point cloud data are selected, and the welding position of the point cloud is extracted by formula (32). The welding position is projected onto the XY plane, and then the two points are connected. The angle between the connecting line and the X-axis is calculated, which is α. Assuming the coordinates of two points are (x_1, y_1, z_1) and (x_2, y_2, z_2), then:

$$\alpha = \arctan\left(\frac{y_2 - y_1}{x_2 - x_1}\right) \tag{36}$$

The calculation of rotation angle is shown in Fig. 11 (b). Similarly, adjacent point cloud data is selected, and the welding position of the point cloud is extracted. The welding position is projected onto the XZ plane, and the angle between the projection and the Y-axis is calculated,

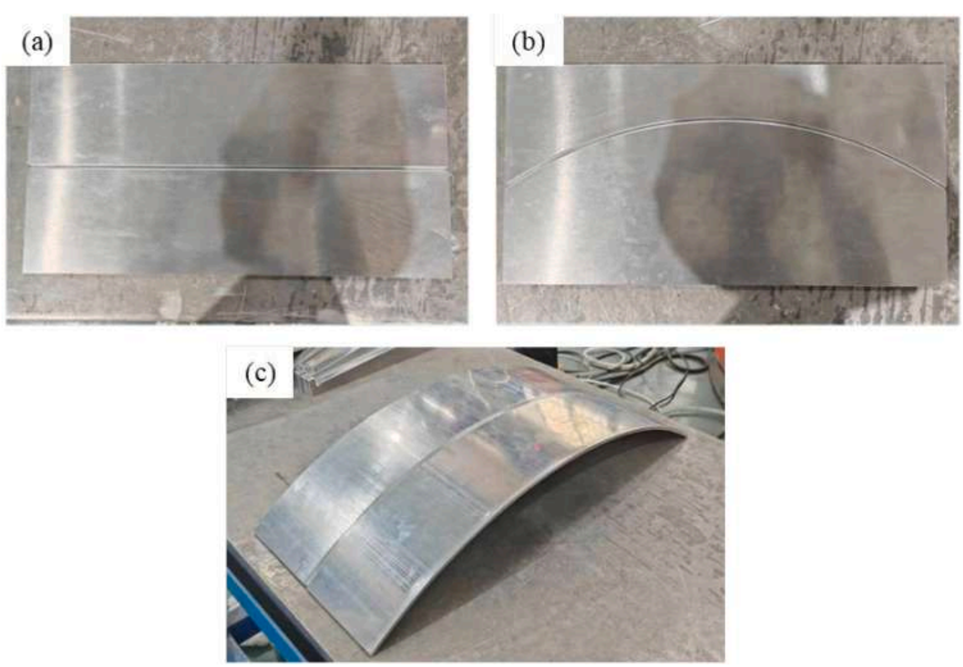

Fig. 12. Workpieces for point cloud construction experiments (a) Linear weld seam (b) Curve weld seam (c) 3D curve weld seam.

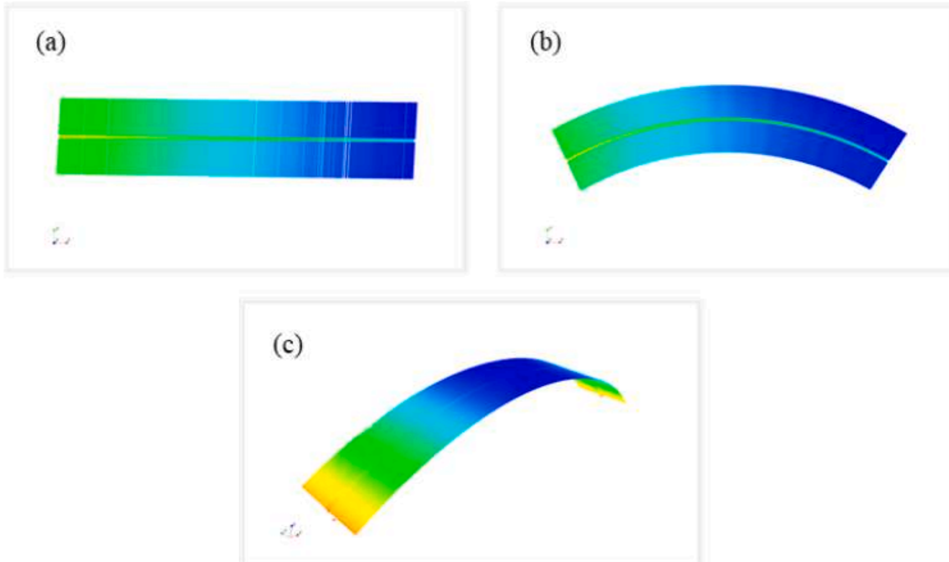

Fig. 13. Point cloud models for various types of workpieces (a) Linear weld seam (b) Curve weld seam (c) 3D curve weld seam.

which is β:

$$\beta = \arctan\left(\frac{z_2 - z_1}{x_2 - x_1}\right) \tag{37}$$

The calculation of the rotation angle is shown in Fig. 11 (c). Similarly, adjacent frames of data are selected and projected onto the YZ plane. Fitting two frames of data into lines, and the angle between two adjacent lines is γ. Assuming the slopes of two lines are k_1 and k_2, then:

$$\gamma = \arctan\left(\frac{k_2 - k_1}{1 + k_1 k_2}\right) \tag{38}$$

Based on angles α, β, and γ, the welding posture adjustment matrix can be calculated:

$$R_{zyx}(\alpha,\beta,\gamma) = \begin{bmatrix} \cos\alpha\cos\beta & \cos\alpha\sin\beta\sin\gamma - \sin\alpha\cos\gamma & \cos\alpha\sin\beta\cos\gamma + \sin\alpha\sin\gamma \\ \sin\alpha\cos\beta & \sin\alpha\sin\beta\sin\gamma + \cos\alpha\cos\gamma & \sin\alpha\sin\beta\cos\gamma - \cos\alpha\sin\gamma \\ -\sin\beta & \cos\beta\sin\gamma & \cos\beta\cos\gamma \end{bmatrix} \tag{39}$$

5. Experiment and discussion

In order to verify the effectiveness of the method proposed in this paper, relevant experiments were designed and conducted, mainly including 3D curve weld point cloud construction experiments, 3D curve weld posture planning experiments, and welding experiments.

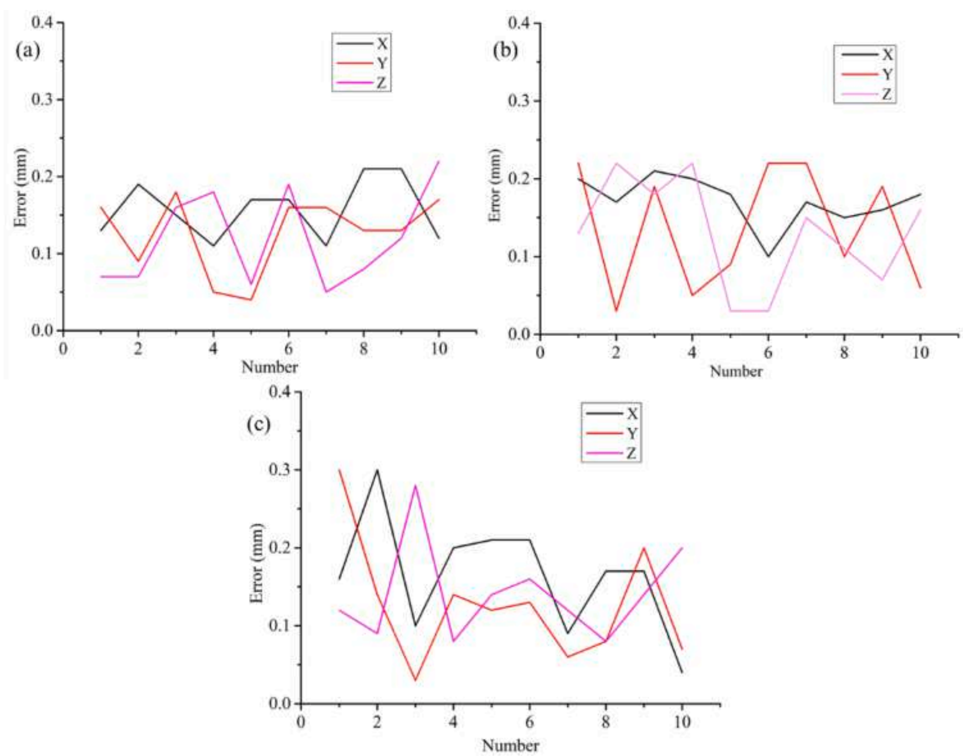

Fig. 14. Error of point cloud (a) Linear weld seam (b) Curve weld seam (c) 3D curve weld seam.

Table 1
Error in point cloud reconstruction experiment.

Type	Distance	Actual value (mm)	Point cloud (mm)	Mean error (mm)
Linear weld seam	ΔX	241	241.17	0.17
	ΔY	177	176.86	0.14
	ΔZ	11	11.13	0.13
Curve weld seam	ΔX	243.6	243.76	0.16
	ΔY	229.7	229.83	0.13
	ΔZ	10.5	10.62	0.12
3D curve weld seam	ΔX	420.8	420.63	0.17
	ΔY	14.7	14.83	0.13
	ΔZ	56.6	56.74	0.14

5.1. Point cloud construction experiment

A point cloud collection experiment was designed, and the experimental workpiece is shown in Fig. 12. In order to comprehensively evaluate the effectiveness of the method, multiple types of workpieces were designed, including linear weld seam, curve weld seam and 3D curve weld seam. The weld seam in this paper is a butt weld seam, and the sensor scanning path is generated by the theoretical model of workpiece. The drawings of each workpiece are analyzed to extract the rough trajectory of the weld seam, which is used as the scanning path for the robot.

The construction of the workpiece point cloud model can be

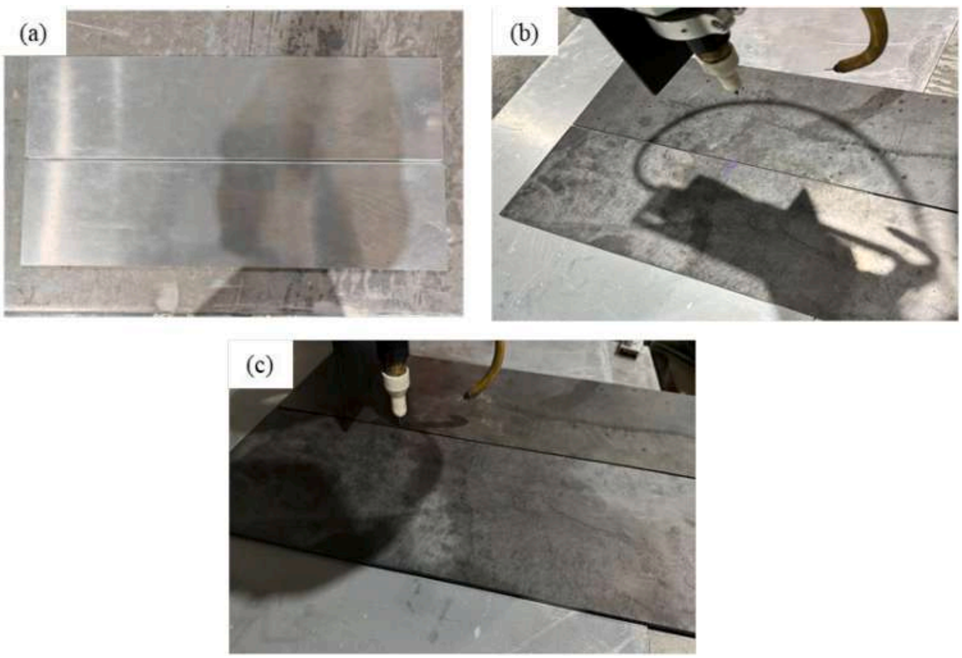

Fig. 15. Point cloud robustness verification experiment (a) Point cloud collection under high brightness and high reflectivity (b) Point cloud collection under high brightness and low reflectivity (c) Point cloud collection under low brightness and low reflectivity.

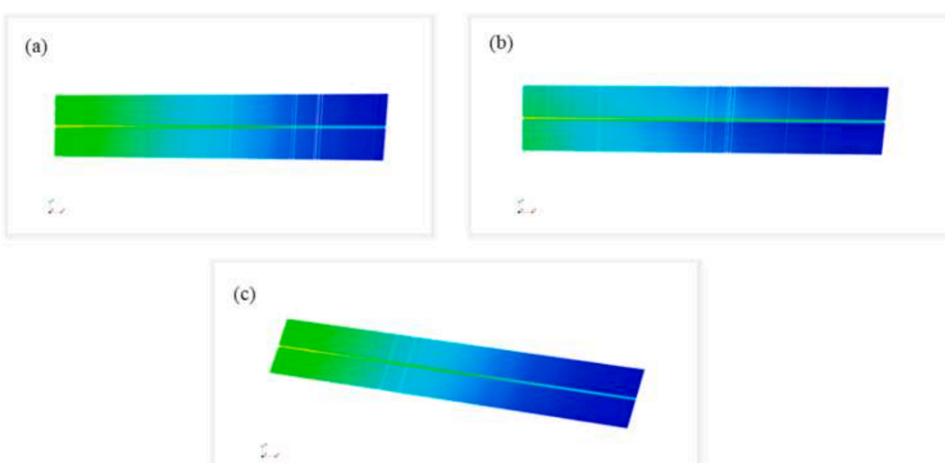

Fig. 16. Point clouds under different light intensities and reflectivity (a) Point clouds under high light intensity and high reflectivity (b) Point cloud collection under high brightness and low reflectivity (c) Point cloud collection under low brightness and low reflectivity.

Table 2
Error of point clouds under different light intensities and reflectivity.

Light intensity	Reflectivity	Average error (mm)
High	high	0.172
High	low	0.158
Low	low	0.152

completed with the method proposed in Section 3. The point clouds of various types of workpieces are shown in Fig. 13. Point clouds are three-dimensional, and in order to make it easier for readers to detect differences in three-dimensional coordinates, point clouds are represented in different colors at different coordinates. It can be seen that the point cloud for linear weld seam, curved weld seam, and 3D curve weld seam have all been constructed, and the point cloud models can represent the actual workpiece. For curved weld seams, the point cloud surface is smooth and continuous without any abrupt changes, which effectively restores the condition of the workpiece.

To further evaluate the accuracy of point clouds, Cloudcompare software is used to measure the distance between the starting and ending points of the point cloud. By comparing with the actual distance, the point cloud error can be obtained. Divide the errors into X, Y, and Z directions, and repeat 10 times to obtain the error of the point cloud. The errors of point cloud reconstruction experiments are shown in Fig. 14. and Table 1. Fig. 14 (a) corresponds to the error of the line weld seam in Fig. 12 (a), Fig. 14 (b) corresponds to the error of the curve weld seam in Fig. 12 (b), and Fig. 14 (c) corresponds to the error of the 3D curve weld seam in Fig. 12 (c). It can be observed that the maximum error of the linear weld seam point cloud in all directions is less than 0.2mm. The maximum error of the 3D curve weld seam point cloud is slightly greater than that of the linear weld seam, but it is still less than 0.3mm. According to Table 1, it can be seen that the average error is less than 0.2mm, which fulfills the practical welding requirements.

Furthermore, experiments were conducted with different reflectivity materials and under different lighting conditions to verify the robustness of the point cloud construction, as shown in Fig. 15. Due to the high reflectivity of aluminum alloy, aluminum alloy and strong light were used in Fig. 15 (a) to create a high brightness and high reflectivity environment. On the other hand, carbon steel has a lower reflectivity. In Fig. 15 (b), carbon steel and strong light were used to create a high brightness and low reflectivity environment. Similarly, in Fig. 15 (c),

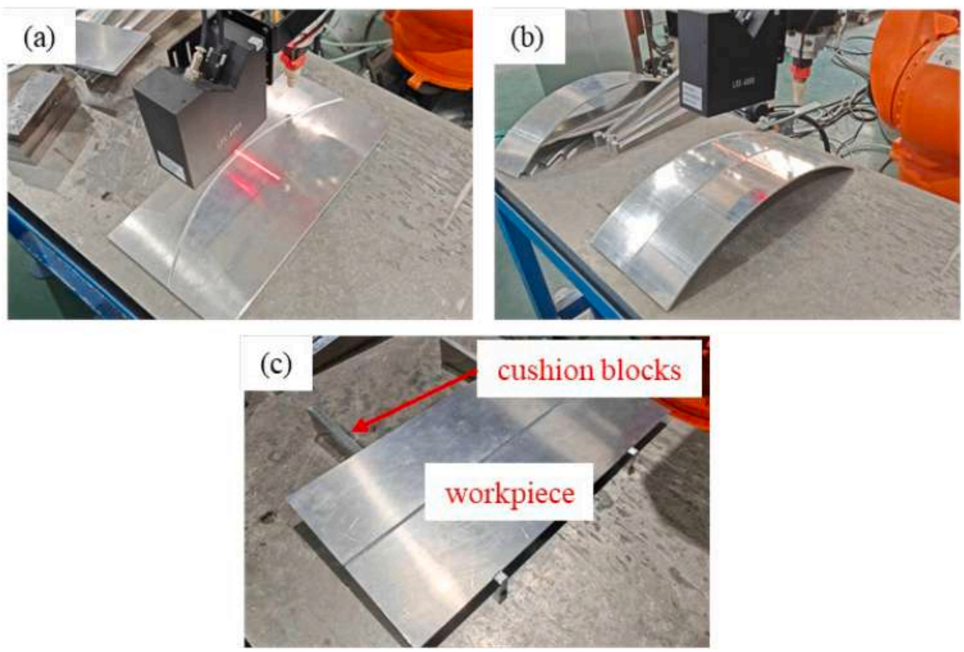

Fig. 17. Posture planning experiment (a) Deflection angle planning (b) Elevation angle planning (c) Rotation angle planning.

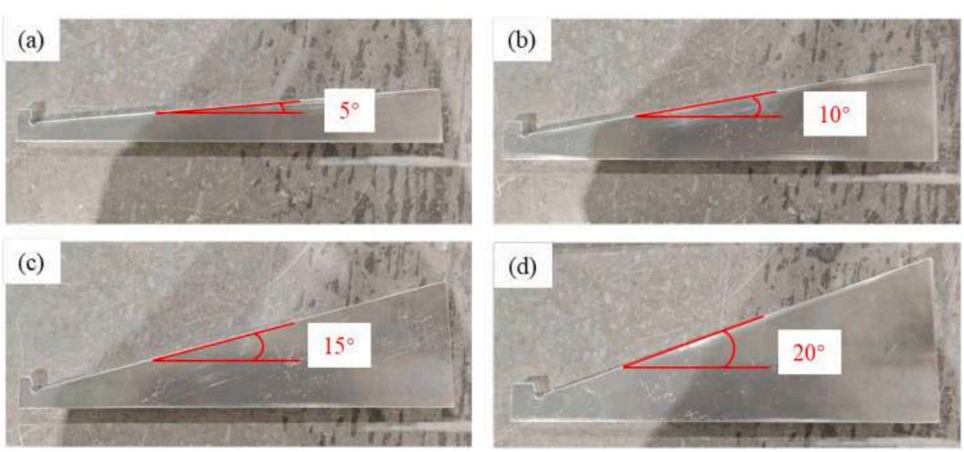

Fig. 18. Multi angle cushion blocks (a)–(d) 5°,10°,15° and 20° cushion blocks.

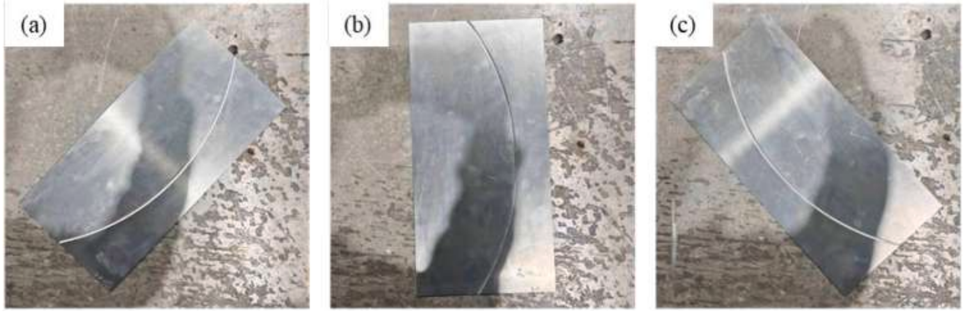

Fig. 19. Deflection angle planning experiment workpiece position (a) Position 1 (b) Position 2 (c) Position 3.

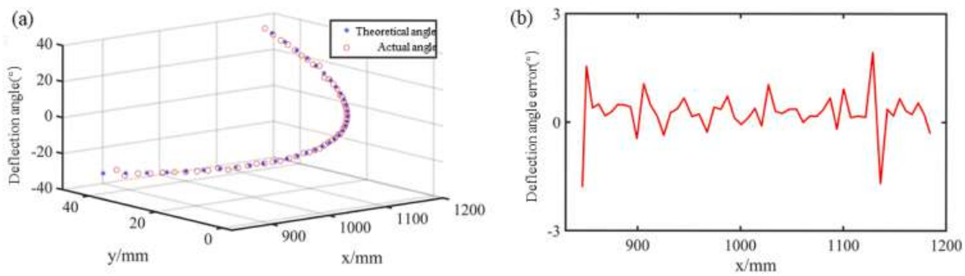

Fig. 20. Deflection angle planning experiment at position 1 (a) Deflection angle (b) Deflection angle error.

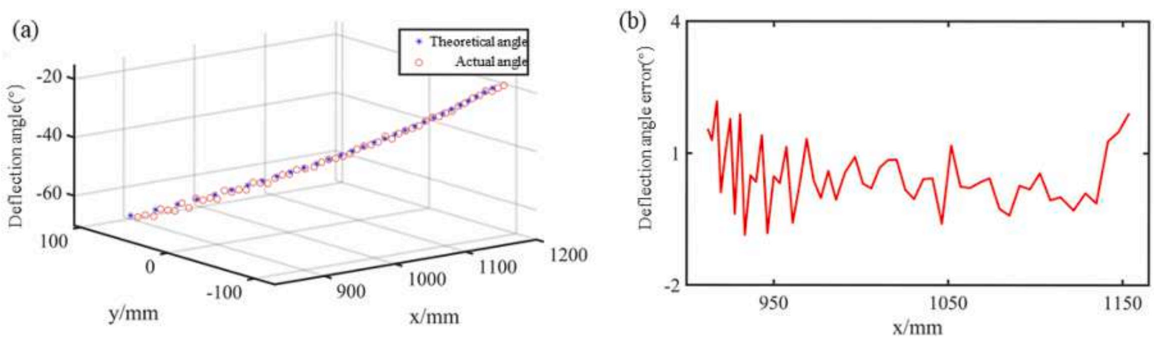

Fig. 21. Deflection angle planning experiment at position 2 (a) Deflection angle (b) Deflection angle error.

carbon steel and low light were used to create a low brightness and low reflectivity environment.

The point clouds collected in the above environment are shown in Fig. 16, and it can be seen that the point clouds collected under different light intensities and reflectivities are similar. Similarly, the error of the point cloud was evaluated, as shown in Table 2. It can be seen that the error under different working conditions is less than 0.2 mm and the fluctuation is small, which shows that the proposed method has good robustness and is sufficient to meet actual needs under various working conditions.

5.2. Curve weld seam posture planning experiment

Due to the welding posture being divided into deflection angle,

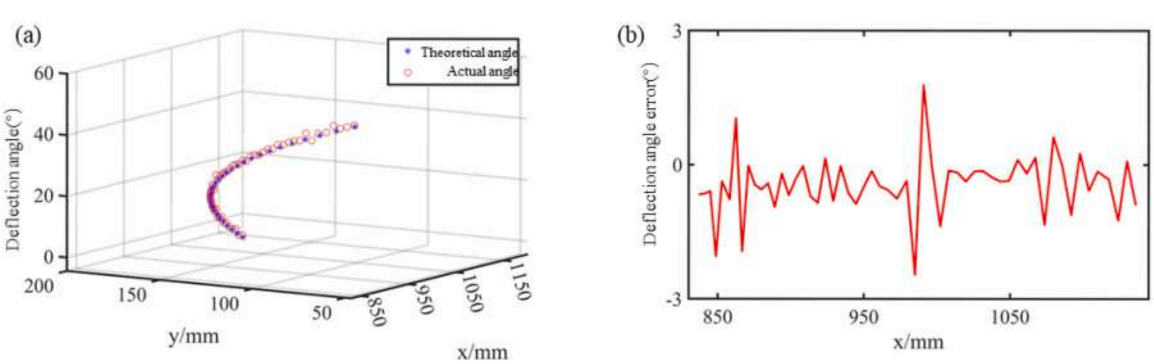

Fig. 22. Deflection angle planning experiment at position 3 (a) Deflection angle (b) Deflection angle error.

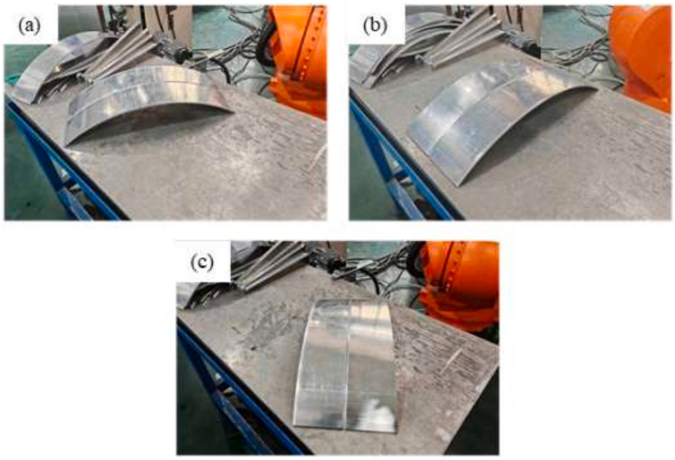

Fig. 23. Elevation angle planning experiment workpiece position (a) Position 1 (b) Position 2 (c) Position 3.

elevation angle, and rotation angle, various posture verification schemes have been designed to accurately verify the planning accuracy of each angle, as shown in Fig. 17. Firstly, the deflection angle planning experiment was conducted. In Fig. 17 (a), the workpiece must be rotated around the Z-axis in TCS during welding to enable the deflection angle planning experiment to be conducted. The second one was the elevation angle planning experiment, as shown in Fig. 17 (b), where the workpiece required the welding torch to be rotated around the Y-axis during welding. The third one was the rotation angle planning experiment, as shown in Fig. 17 (c). To validate the effectiveness of different angle planning, multiple cushion blocks with different angles were designed, as shown in Fig. 18. Placing the workpiece on the cushion blocks enables the completion of a multi rotation angle planning experiment.

5.2.1. Deflection angle planning

In this experiment, the workpiece was placed in three different positions, as shown in Fig. 19. Experiments on deflection angle planning at positions 1, 2, and 3 were then conducted. Utilizing the methods in Section 4, the deflection angle planning can be completed. To evaluate the accuracy of the planned posture, a comparison was conducted between the planned angle and the theoretically optimal angle.

The planned deflection angles and errors for each position are shown in Fig. 20–Fig. 22. Fig. 20 shows the planning angle and error corresponding to position 1, Fig. 21 shows the planning angle and error corresponding to position 2, and Fig. 22 shows the planning angle and error corresponding to position 3. It can be observed that in all three positions, the planned angle corresponds closely with the theoretical optimal angle, with a maximum error angle of less than 2.2° and an average error of about 0.75°, which can meet the demand of practical welding operations..

5.2.2. Elevation angle planning

The position of the workpiece in this experiment is shown in Fig. 23. The experiments on elevation angle planning at positions 1, 2, and 3 were conducted. The angles between positions 1, 2, and 3 and the coordinate axis are 40 °, 90 °, and 135 °, respectively. Similar to the previous experiment, the planning angle was also compared with the theoretical optimal angle to evaluate the planning accuracy.

The results of elevation angle planning are shown in Figs. 24–26. Fig. 24 shows the planning angle and error corresponding to position 1, Fig. 25 shows the planning angle and error corresponding to position 2, and Fig. 26 shows the planning angle and error corresponding to position 3. It can be found that the planning angle corresponds closely with the theoretical optimal angle, with a maximum error of less than 3 ° and an average error of 1.2 °, which satisfies the practical welding needs.

5.2.3. Rotation angle planning

The rotation angle planning experiment was conducted with four different angle pads as shown in Fig. 18, and the experimental

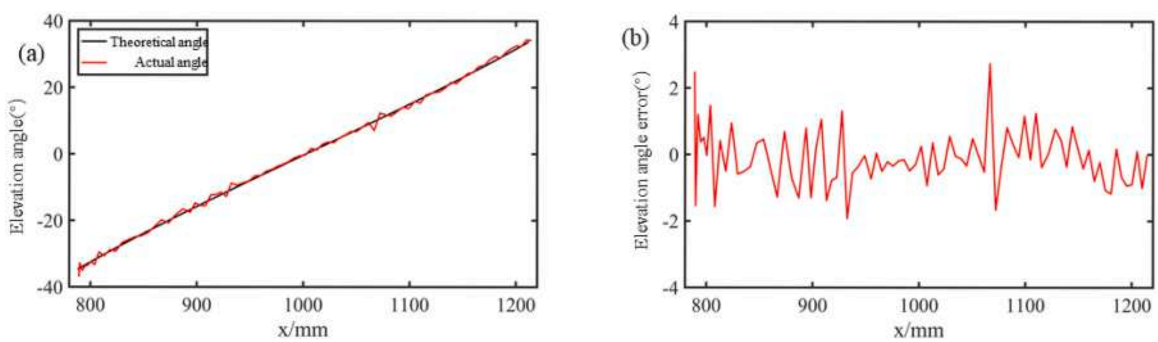

Fig. 24. Elevation angle planning experiment at position 1 (a) Elevation angle (b) Elevation angle error.

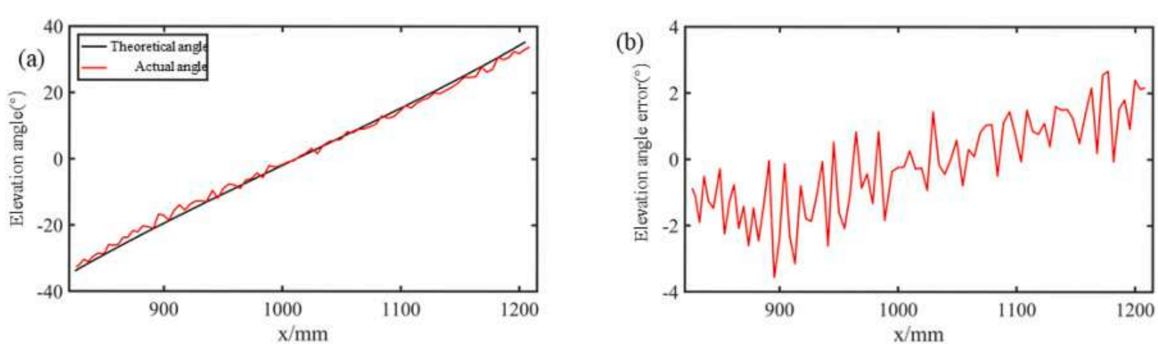

Fig. 25. Elevation angle planning experiment at position 2 (a) Elevation angle (b) Elevation angle error.

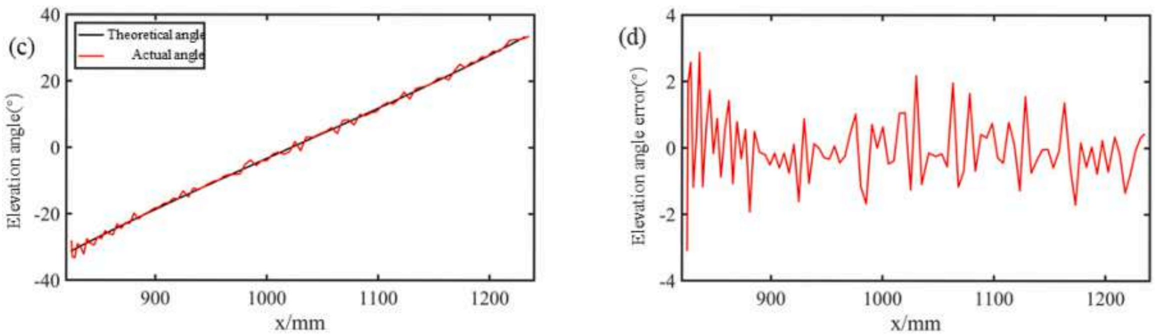

Fig. 26. Elevation angle planning experiment at position 3 (a) Elevation angle (b) Elevation angle error.

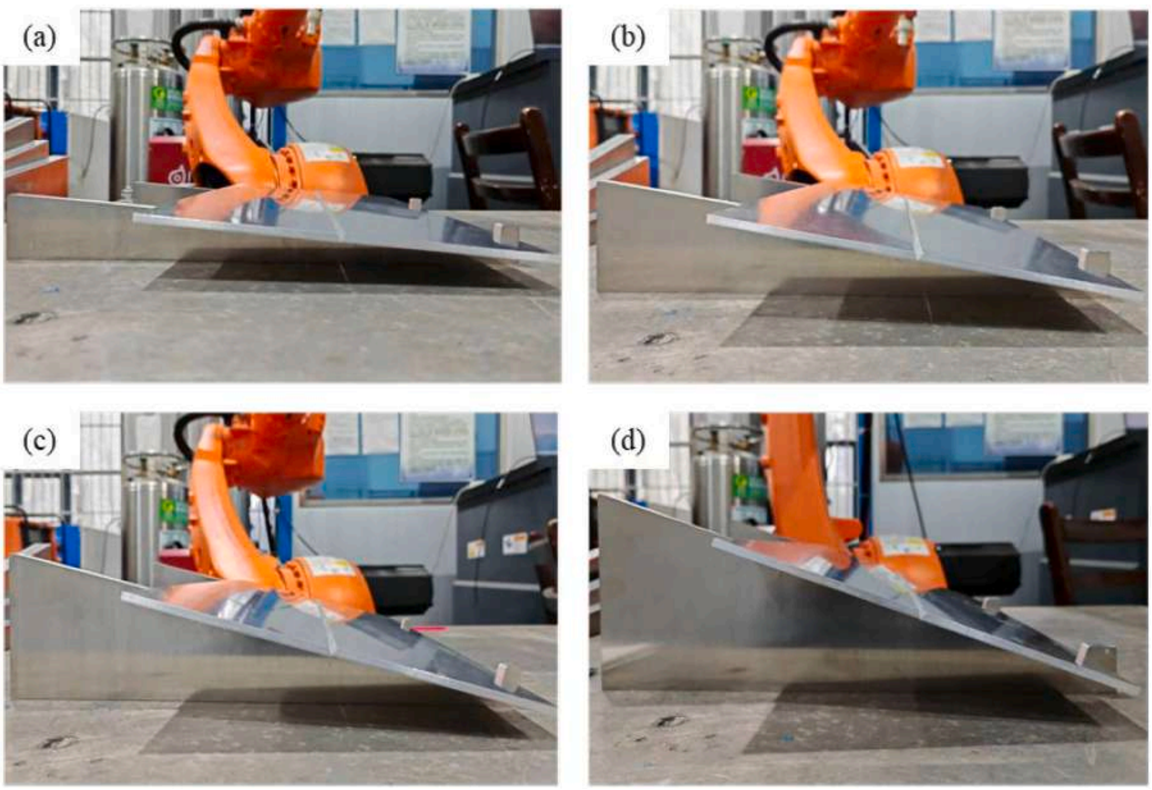

Fig. 27. Rotation angle planning experiment workpiece position (a) 5° cushion block (b) 10° cushion block (c) 15° cushion block (d) 20° cushion block.

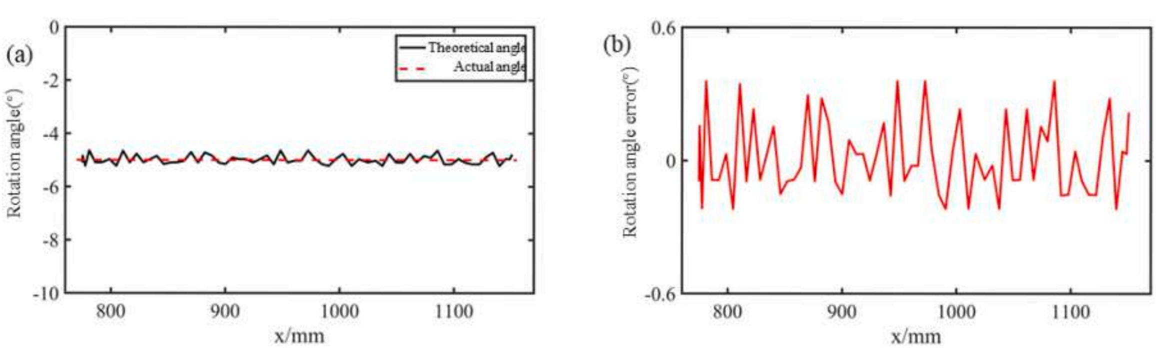

Fig. 28. Rotation angle planning experiment with 5° cushion block (a) Rotation angle (b) Rotation angle error.

workpiece arrangement is shown in Fig. 27, corresponding to 5°, 10°, 15°, and 20°. The results and errors of the rotation angle planning are shown in Fig. 27–31. Fig. 28 shows the experimental results with the 5° cushion block, Fig. 29 shows the experimental results with the 10° cushion block, Fig. 30 shows the experimental results with the 15° cushion block, and Fig. 31 shows the experimental results with the 20° cushion block. It can be seen that the planned angle of rotation corresponds closely to the actual angle, with a maximum error of less than 0.6° and an average error of 0.28°, which meets the practical welding needs.

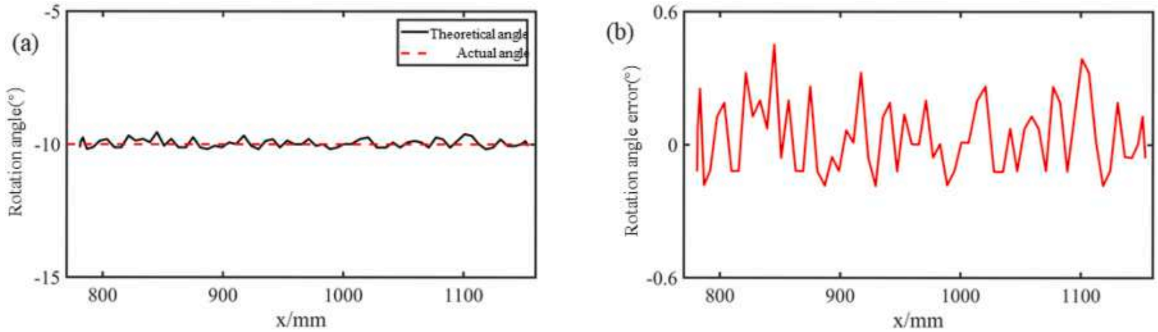

Fig. 29. Rotation angle planning experiment with 10° cushion block (a) Rotation angle (b) Rotation angle error.

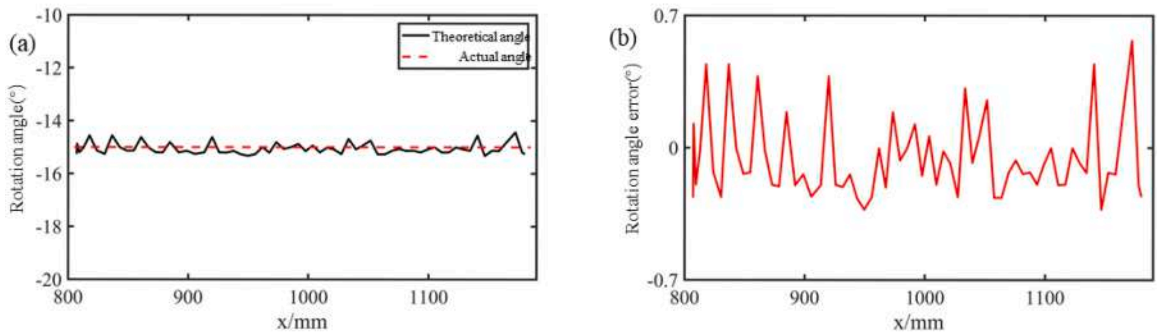

Fig. 30. Rotation angle planning experiment with 15° cushion block (a) Rotation angle (b) Rotation angle error.

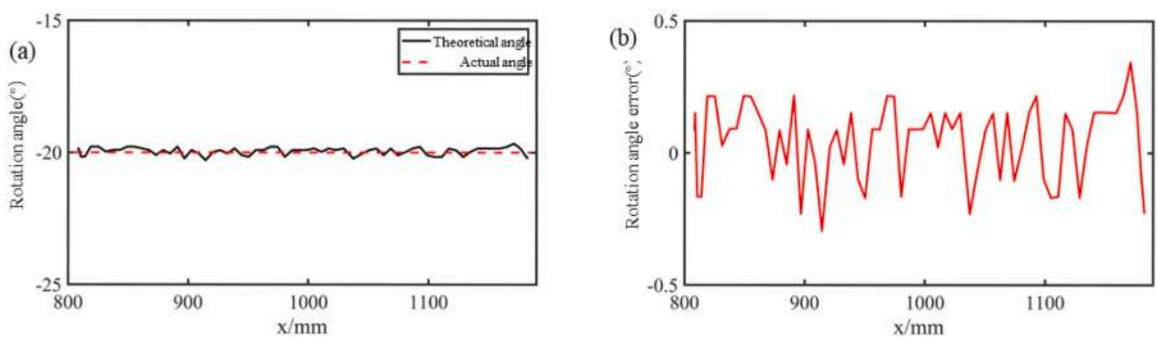

Fig. 31. Rotation angle planning experiment with 20° cushion block (a) Rotation angle (b) Rotation angle error.

Table 3
The algorithm running time.

Key steps	Running time (ms)
Point cloud construction (one frame)	1.5
Line extraction (one frame)	1.6
Path planning (one frame)	2.8
Posture planning (one step)	2.4

5.2.4. Efficiency of the proposed method

In order to comprehensively evaluate the performance of the proposed method, the efficiency of the algorithm was tested. The CPU of the computer used in this experiment is Intel(R) Core(TM) i7–9750H. Different point cloud models have different sizes, resulting in different algorithm operation times. In order to rigorously evaluate the efficiency of the algorithm, the operation time of a single frame is calculated. The running time of each key step is shown in Table 3. It can be seen that the running time of each step is between 1.5ms and 3ms, which proves the high efficiency of the proposed method.

5.2.5. Comparison with the state of art methods

To comprehensively evaluate the effectiveness of the method proposed in the paper, a comparison was conducted with the state of art methods. Papers [34] and [35] also estimated the welding posture with different methods. Among them, a visual sensor was used in paper [34] to collect and process weld seam images and then complete posture calculation, while a line laser sensor was used in paper [35] to process the weld point cloud and complete welding posture calculation.

The comparison results between the current research and the methods presented in the two papers are shown in Table 4. It can be

Table 4
Comparison with the state of art methods.

Method	Posture	Average error°	Maximum error°
The proposed method	Deflection angle	0.75	2.18
	Elevation angle	1.20	2.87
	Rotation angle	0.28	0.57
Method in [34]	Deflection angle	1.08	2.35
	Elevation angle	1.23	2.44
	Rotation angle	0.43	1.50
Method in [35]	Deflection angle	1.60	1.90
	Elevation angle	1.40	1.80
	Rotation angle	/	/

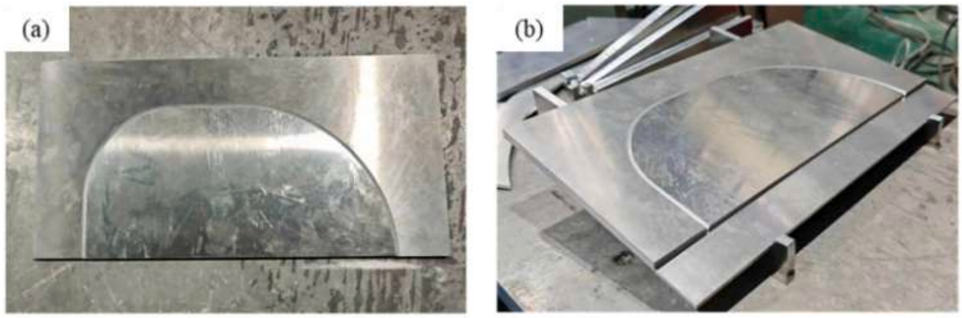

Fig. 32. Welding verification experiment workpiece (a) The shape of the workpiece (b) The posture of the workpiece.

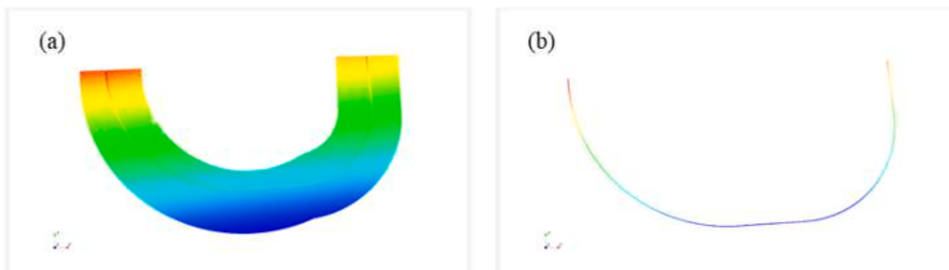

Fig. 33. Experimental workpiece point cloud construction and weld path extraction (a) Point cloud of workpiece (b) Extracted weld seam path.

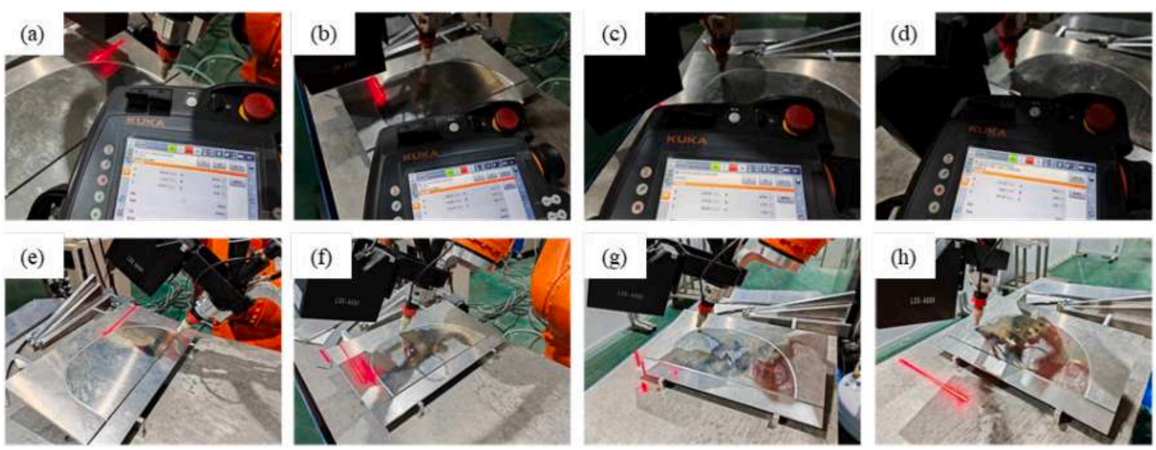

Fig. 34. The effect of posture planning (a)–(d) Welding torch posture before planning (e)–(h) Welding torch posture after planning.

observed that the method proposed in this paper reduces the average deflection angle error by 31 % and 53 % compared to the two methods. The average error of the elevation angle is slightly reduced and the maximum error is slightly increased. The average and maximum errors of the rotation angle are also reduced, with an average error reduction of 35 %, while the calculation of the rotation angle was not performed in paper [35]. In summary, by considering the three attitude angles, the posture planning effect has been significantly improved.

5.3. Welding verification experiment

To further test the effectiveness of point cloud construction and posture planning in this paper, welding validation experiments were conducted in this experiment. Considering that the posture angle is divided into deflection angle, elevation angle, and rotation angle, a special workpiece was designed in this experiment, as shown in Fig. 32, which contains line welds and curved welds with multiple curvatures. The workpiece was placed on the 10° cushion block during welding. When conducting welding experiments, this workpiece must adjust three posture angles simultaneously, so the effectiveness of the method proposed in this paper can be tested.

Firstly, the point cloud for the workpiece was constructed, as shown in Fig. 33 (a). Then, the weld path was extracted, as shown in Fig. 33 (b), it can be seen that the weld seam has been accurately extracted. Furthermore, the welding posture was planned, as shown in Fig. 34. Comparing the angles before and after planning, it can be observed that the welding torch posture changes at each position of the workpiece to adapt to curved weld seams. The obtained welding effect of the workpiece is shown in Fig. 35. It can be seen that the weld seam is uniform and complete, and the welding effect meets the practical requirements.

6. Conclusion

In this paper, a point cloud construction method based on robot pose is proposed for 3D curved weld seams. Furthermore, the posture of the welding torch for 3D curve weld seams has been planned. The main

Fig. 35. Obtained welding effect of curved weld seams.

contents of this paper include:

(1) Given the multiple coordinate systems of the welding robot and line laser sensor, and the conversion relationship between multiple coordinate systems is accurately solved, laying the solid foundation for subsequent research.
(2) This paper proposes a point cloud construction method based on robot pose and line laser sensor. A single frame point cloud is obtained through line laser sensor, and coordinate transformation between multiple frame point clouds is achieved through robot pose to complete the construction of complex weld point cloud models. Experimental results demonstrate that the point cloud construction method proposed in this paper has an accuracy of better than 0.2 mm.
(3) A based point cloud processing method based on the RANSAC algorithm is proposed to extract centerline information from point clouds. Furthermore, welding position calculation is completed through a common perpendicular line.
(4) Based on the motion characteristics of welding robots, the welding posture is divided into deflection angle, elevation angle, and rotation angle. Additionally, methods for calculating these posture angles have been proposed. Experimental results have shown that the three angle errors are 0.75°, 1.2°, and 0.28°, which represents a significant improvement compared to the state of art methods and aligns well with the practical welding operations.

However, this paper mainly focuses on butt welds, and multiple types of welds such as lap joints and T-joints are still under study. In the future, research will be conducted on multiple types of weld joints to expand the application scope of the proposed method. Furthermore, deep learning methods will be considered to improve the robustness and efficiency.

CRediT authorship contribution statement

Hui Wang: Writing – original draft, Methodology, Writing – review & editing. **Youmin Rong:** Writing – review & editing, Conceptualization. **Songming Xiang:** Data curation, Software. **Jiajun Xu:** Writing – review & editing. **Yifan Peng:** Investigation. **Yu Huang:** Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Disclosure statement

No potential conflict of interest was reported by the authors.

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Data availability

The authors are unable or have chosen not to specify which data has been used.

References

[1] D. Lee, N. Ku, T.W. Kim, J. Kim, K.Y. Lee, Y.S. Son, Development and application of an intelligent welding robot system for shipbuilding, Robot. Comput. Integr. Manuf. 27 (2011) 377–388, https://doi.org/10.1016/j.rcim.2010.08.006.
[2] G. Shang, L. Xu, Z. Li, Z. Zhou, Z. Xu, Digital-twin-based predictive compensation control strategy for seam tracking in steel sheets welding of large cruise ships, Robot. Comput. Integr. Manuf. 88 (2024) 102725, https://doi.org/10.1016/j.rcim.2024.102725.
[3] L.J. Jia, H. Ge, T. Suzuki, X. Luo, Experimental study on cracking of thick-walled welded beam-column connections with incomplete penetration in steel bridge piers, J. Bridg. Eng. (2015) 20, https://doi.org/10.1061/(ASCE)BE.1943-5592.0000659.
[4] D. Yuan, C. Cui, Q. Zhang, Z. Li, Z. Ye, Fatigue damage evaluation of welded joints in steel bridge based on meso-damage mechanics, Int. J. Fatigue. 161 (2022) 106898, https://doi.org/10.1016/j.ijfatigue.2022.106898.
[5] G. Alencar, A. de Jesus, J.G.S. da Silva, R. Calçada, Fatigue cracking of welded railway bridges: A review, Eng. Fail. Anal. 104 (2019) 154–176, https://doi.org/10.1016/j.engfailanal.2019.05.037.
[6] Q. Feng, R. Li, B. Nie, S. Liu, L. Zhao, H. Zhang, Literature review: theory and application of In-line inspection technologies for oil and gas pipeline girth weld defection, SENSORS (2017) 17, https://doi.org/10.3390/s17010050.
[7] H. Li, L. Li, X. Chen, Y. Zhou, Z. Li, Z. Zhao, Addressing the inspection selection challenges of in-service pipeline weld ensemble tree models, Eng. Fail. Anal. (2024) 156, https://doi.org/10.1016/j.engfailanal.2023.107852.
[8] S.K. Sharma, S. Maheshwari, A review on welding of high strength oil and gas pipeline steels, J. Nat. GAS Sci. Eng. 38 (2017) 203–217, https://doi.org/10.1016/j.jngse.2016.12.039.
[9] Y. Gao, P. Wang, K. Wang, J. Xu, Z. Dong, Damage tolerance of fractured rails on continuous welded rail track for high-speed railways, Railw. Eng. Sci. 29 (2021) 59–73, https://doi.org/10.1007/s40534-020-00226-7.
[10] S. Wang, Z. Xu, C. Wu, L. Hua, D. Zhu, Towards region-based robotic machining system from perspective of intelligent manufacturing: A technology framework with case study, J. Manuf. Syst. 70 (2023) 451–463, https://doi.org/10.1016/j.jmsy.2023.08.017.
[11] M.S. Khan, M. Soleimani, A.R.H. Midawi, I. Aderibigbe, Y.N. Zhou, E. Biro, A review on heat affected zone softening of dual-phase steels during laser welding, J. Manuf. Process. 102 (2023) 663–684, https://doi.org/10.1016/j.jmapro.2023.07.059.
[12] H. Li, B. Li, G. Liu, X. Wen, H. Wang, X. Wang, S. Zhang, Z. Zhai, W. Yang, A detection and configuration method for welding completeness in the automotive body-in-white panel based on digital twin, Sci. Rep. (2022) 12, https://doi.org/10.1038/s41598-022-11440-0.
[13] F. Xu, Y. Xu, H. Zhang, S. Chen, Application of sensing technology in intelligent robotic arc welding: A review, J. Manuf. Process. 79 (2022) 854–880, https://doi.org/10.1016/j.jmapro.2022.05.029.
[14] M. Rodríguez-Martín, P. Rodríguez-Gonzálvez, S. Lagüela, D. González-Aguilera, Macro-photogrammetry as a tool for the accurate measurement of three-dimensional misalignment in welding, Autom. Constr. 71 (2016) 189–197, https://doi.org/10.1016/j.autcon.2016.08.016.
[15] O. Jing, Z. Lai, W. Qinghong, L. Xin, L. Yingjie, Weld-seam identification and model reconstruction of remanufacturing blade based on three-dimensional vision, Adv. Eng. Informatics. 49 (2021) 101300, https://doi.org/10.1016/j.aei.2021.101300.
[16] L.D. Hanh, L.D. Dao, N.C. Luan, 3D complex curve seam tracking using industrial robot based on CAD model and computer vision, Int. J. Interact. Des. Manuf. 17 (2023) 1039–1046, https://doi.org/10.1007/s12008-022-01043-4.
[17] Z. Tan, B. Zhao, Y. Ji, X. Xu, Z. Kong, T. Liu, M. Luo, A welding seam positioning method based on polarization 3D reconstruction and linear structured light imaging, Opt. Laser Technol. 151 (2022) 108046, https://doi.org/10.1016/j.optlastec.2022.108046.
[18] B. Li, Z. Xu, F. Gao, Y. Cao, Q. Dong, 3D Reconstruction of high reflective welding surface based on binocular structured light stereo vision, machines. 10 (2022). https://doi.org/10.3390/machines10020159.

[19] P. Chi, Z. Wang, H. Liao, T. Li, J. Zhan, X. Wu, J. Tian, Q. Zhang, Low-latency visual-based high-quality 3-D reconstruction using point Cloud optimization, IEEE Sens. J. 23 (2023) 20055–20065, https://doi.org/10.1109/JSEN.2023.3297232.
[20] X. Wang, X. Chen, Z. Zhao, Y. Zhang, D. Zheng, J. Han, High-precision point cloud registration system of multi-view industrial self-similar workpiece based on super-point space guidance, J. Intell. Manuf. (2023), https://doi.org/10.1007/s10845-023-02136-x.
[21] H. Wang, Y. Huang, G. Zhang, Y. Rong, A novel method for dense point cloud reconstruction and weld seam detection for tubesheet welding robot, Opt. LASER Technol. 163 (2023) 109346, https://doi.org/10.1016/j.optlastec.2023.109346.
[22] H. Wang, Y. Rong, C. Liu, Y. Huang, Construction of a semi-dense point cloud model for a tube-to-tubesheet welding robot, IET Collab. Intell. Manuf. (2022), https://doi.org/10.1049/cim2.12056.
[23] R. Xiao, Y. Xu, Z. Hou, C. Chen, S. Chen, An adaptive feature extraction algorithm for multiple typical seam tracking based on vision sensor in robotic arc welding, Sensors Actuators, A Phys 297 (2019) 111533, https://doi.org/10.1016/j.sna.2019.111533.
[24] H. Wang, Y.M. Rong, J.J. Xu, S.M. Xiang, Y.F. Peng, Y. Huang, WeldNet: A voxel-based deep learning network for point cloud annular weld seam detection, Sci. China Technol. Sci. 67 (2024) 1215–1225, https://doi.org/10.1007/s11431-023-2569-1.
[25] W. Fang, X. Xu, X. Tian, A vision-based method for narrow weld trajectory recognition of arc welding robots, Int. J. Adv. Manuf. Technol. 121 (2022) 8039–8050, https://doi.org/10.1007/s00170-022-09804-x.
[26] L. Yang, E. Li, T. Long, J. Fan, Z. Liang, A novel 3-D path extraction method for arc welding robot based on stereo structured light sensor, IEEE Sens. J. 19 (2019) 763–773, https://doi.org/10.1109/JSEN.2018.2877976.
[27] Q. Wu, Z. Li, C. Gao, W. Biao, G. Shen, Research on welding guidance system of intelligent perception for steel weldment, IEEE Sens. J. 23 (2023) 5220–5231, https://doi.org/10.1109/JSEN.2023.3235454.
[28] L. Yang, Y. Liu, J. Peng, Z. Liang, A novel system for off-line 3D seam extraction and path planning based on point cloud segmentation for arc welding robot, Robot. Comput. Integr. Manuf. 64 (2020) 101929, https://doi.org/10.1016/j.rcim.2019.101929.
[29] Y. Geng, M. Lai, X. Tian, X. Xu, Y. Jiang, Y. Zhang, A novel seam extraction and path planning method for robotic welding of medium-thickness plate structural parts based on 3D vision, Robot. Comput. Integr. Manuf. (2023) 79, https://doi.org/10.1016/j.rcim.2022.102433.
[30] Y. Zou, R. Zhan, Research on 3D curved weld seam trajectory position and orientation detection method, Opt. Lasers Eng. 162 (2023) 107435, https://doi.org/10.1016/j.optlaseng.2022.107435.
[31] E. Pedrosa, M. Oliveira, N. Lau, V. Santos, A general approach to hand-eye calibration through the optimization of atomic transformations, IEEE Trans. Robot. 37 (2021) 1619–1633, https://doi.org/10.1109/TRO.2021.3062306.
[32] F. Liang, C. Kang, F. Fang, A smooth tool path planning method on NURBS surface based on the shortest boundary geodesic map, J. Manuf. Process. 58 (2020) 646–658, https://doi.org/10.1016/j.jmapro.2020.08.047.
[33] S. Zhao, F. Peng, H. Sun, R. Yan, X. Tang, H. Zhang, Robotic milling posture adjustment under composite constraints: A weight-sequence identification and optimization strategy, Robot. Comput. Integr. Manuf. 85 (2024) 102635, https://doi.org/10.1016/j.rcim.2023.102635.
[34] Y. Zou, J. Chen, X. Wei, Research on a real-time pose estimation method for a seam tracking system, Opt. Lasers Eng. 127 (2020) 105947, https://doi.org/10.1016/j.optlaseng.2019.105947.
[35] B. Hong, A. Jia, Y. Hong, X. Li, J. Gao, Y. Qu, Online extraction of pose information of 3d zigzag-line welding seams for welding seam tracking, Sensors (Switzerland) 21 (2021) 1–19, https://doi.org/10.3390/s21020375.